

PERCENTAGES — Short Revision Notes

Evaluate Level: Make Judgments and Justify Decisions

1 Verifying Whether an Answer Is Correct

Problem	Student's Answer	Verification	Judgment
CP=500, SP=650. Find Profit%	30%	Profit=150, $150/500 \times 100 = 30\%$	CORRECT
CP=500, SP=450. Find Loss%	10%	Loss=50, $50/500 \times 100 = 10\%$	CORRECT
CP=800, Profit=20%. Find SP	960	$800 \times 120/100 = 960$	CORRECT
CP=800, Loss=10%. Find SP	720	$800 \times 90/100 = 720$	CORRECT
MP=1000, Discount=20%. Find SP	800	$1000 \times 80/100 = 800$	CORRECT
CP=600, SP=750. Find Profit%	20%	Profit=150, $150/600 \times 100 = 25\%$	INCORRECT
CP=1000, Loss=20%. Find SP	800	$1000 \times 80/100 = 800$	CORRECT

Verification Method:

1. Identify what is given (CP, SP, MP, etc.)
2. Apply the correct formula
3. Calculate carefully
4. Compare with student's answer

2 Judging the Correct Formula

Scenario	Formula Used	Correct?	Why / Why Not
Finding Profit%	$(\text{Profit} \div \text{SP}) \times 100$	WRONG	Should use CP, not SP
Finding Profit%	$(\text{Profit} \div \text{CP}) \times 100$	CORRECT	Profit% is based on CP
Finding Loss%	$(\text{Loss} \div \text{SP}) \times 100$	WRONG	Should use CP, not SP
Finding Loss%	$(\text{Loss} \div \text{CP}) \times 100$	CORRECT	Loss% is based on CP
Finding SP with Profit	$\text{CP} \times (100 + P\%)/100$	CORRECT	Correct formula
Finding SP with Loss	$\text{CP} \times (100 - L\%)/100$	CORRECT	Correct formula

Common Mistakes to Judge

Mistake	Why Wrong	Correct Version
Profit% = Profit/SP	Profit% uses CP as base	$\text{Profit}/\text{CP} \times 100$
Loss% = Loss/SP	Loss% uses CP as base	$\text{Loss}/\text{CP} \times 100$
Discount% = Discount/CP	Discount uses MP as base	$\text{Discount}/\text{MP} \times 100$
Commission% = Commission/CP	Commission uses Value as base	$\text{Commission}/\text{Value} \times 100$
Markup% = Markup/SP	Markup uses CP as base	$\text{Markup}/\text{CP} \times 100$

3 Evaluating Reasonableness

Is the Answer Reasonable?

Scenario	Answer	Reasonable?	Justification
CP=100, Profit=10%	SP=110	YES	10% of 100 = 10, SP=110
CP=100, Profit=100%	SP=200	YES	Double the CP
CP=100, Profit=200%	SP=300	YES	100 + 200 = 300
CP=100, Loss=10%	SP=90	YES	100 - 10 = 90
CP=100, Loss=100%	SP=0	YES	100 - 100 = 0
CP=100, Profit=20%	SP=100	NO	Should be 120, not 100
CP=100, Loss=20%	SP=120	NO	Loss means SP should be LESS
MP=1000, Disc=10%	SP=1100	NO	Discount means SP should be LESS

Reasonableness Checklist:

- If there is profit → SP must be > CP
- If there is loss → SP must be < CP
- If there is discount → SP must be < MP
- If there is commission → Net must be < Value

4 Judging the Best Deal

Comparing Discounts

Option	Deal	Calculation	You Pay	Evaluation
Shop A	8% discount on Rs 1500	$1500 \times 92\% = 1380$	Rs 1380	Better
Shop B	Rs 100 off on Rs 1500	$1500 - 100 = 1400$	Rs 1400	Worse

Justification: Shop A saves Rs 120, Shop B saves Rs 100. Shop A is better.

Comparing More Deals

Option	Deal	You Pay	Evaluation
Shop A	10% discount	90% of price	—
Shop B	Rs 50 off on purchases over Rs 500	Price - 50	—

Price	Shop A (10%)	Shop B (Rs 50 off)	Better
400	360	350	Shop B
500	450	450	Equal
600	540	550	Shop A
1000	900	950	Shop A

Conclusion:

- Below Rs 500 → Shop B is better
- Above Rs 500 → Shop A is better
- At Rs 500 → Both are equal

5 Evaluating Profitability**Which Is More Profitable?**

Scenario A	Scenario B	Evaluation
CP=800, SP=900	CP=2500, SP=2600	—
Profit=100	Profit=100	
Profit%=12.5%	Profit%=4%	

Judgment: Scenario A is more profitable (12.5% > 4%)

Scenario A	Scenario B	Evaluation
CP=800, Profit=20%	CP=2500, Profit=20%	—
Profit=160	Profit=500	

Judgment: Scenario B earns more absolute profit (500 > 160)

Evaluation Criteria:

- If considering efficiency → Higher Profit% is better
- If considering actual earnings → Higher absolute Profit is better

6 Judging Student Mistakes

Student's Work	Analysis	Judgment	Correction
CP=800, Profit=25%. SP = $800 + 25 = 825$	Added 25 instead of 25%	WRONG	$800 \times 125/100 = 1000$
MP=1200, Discount=20%. SP = $1200 - 20 = 1180$	Subtracted 20 instead of 20%	WRONG	$1200 \times 80/100 = 960$
CP=600, SP=900. Profit% = $300/900 \times 100 = 33.3\%$	Used SP instead of CP	WRONG	$300/600 \times 100 = 50\%$
MP=1000, SP=800. Discount% = $200/800 \times 100 = 25\%$	Used SP instead of MP	WRONG	$200/1000 \times 100 = 20\%$
SP=550, Profit=10%. CP = $550 \times 110/100 = 605$	Used wrong formula	WRONG	$550 \times 100/110 = 500$

Common Error Types

Error Type	Example	Correct
Mistaking percentage for amount	$25\% = 25$	$25\% = 25/100$
Using wrong base	Profit% = Profit/SP	Profit% = Profit/CP
Wrong formula direction	$CP = SP \times (100+P)/100$	$CP = SP \times 100/(100+P)$
Adding percentages incorrectly	$20\% + 20\% = 40\%$	Need step-by-step calculation

7 Evaluating the Seller's Decision

Seller	CP	MP	Discount %	SP	Profit/Loss	Evaluation
A	100	120	10%	108	Profit 8%	Good
B	100	120	20%	96	Loss 4%	Bad
C	100	125	20%	100	Break-even	Neutral
D	100	150	20%	120	Profit 20%	Best

A: $MP = CP + 20\%$, Discount = 10% → Profit = 8% (Good but not best)
 B: $MP = CP + 20\%$, Discount = 20% → Loss = 4% (Bad decision)
 C: $MP = CP + 25\%$, Discount = 20% → Break-even (No profit)
 D: $MP = CP + 50\%$, Discount = 20% → Profit = 20% (Best decision)

Judgment Criteria:

- Is $SP > CP$? → Profit
- Is $SP < CP$? → Loss
- Is $SP = CP$? → Break-even
- Higher profit% is better for seller

8 Evaluating Customer Decisions

Which Discount Is Better for the Customer?

MP	Option A: 10% discount	Option B: Rs 100 off	Evaluation
1000	You pay 900	You pay 900	Equal!
800	You pay 720	You pay 700	Option B better
1200	You pay 1080	You pay 1100	Option A better

Evaluation Rules: Compare the actual amount you pay — lower amount = better for customer.

- At $MP = 800$: $10\% = 80$, Rs 100 off = 100 → Rs 100 off is better
- At $MP = 1200$: $10\% = 120$, Rs 100 off = 100 → 10% is better

9 Evaluating Multi-Step Transactions

Scenario: Carpenter → Vendor → Customer

Step	CP	Profit %	SP
Carpenter	1800	20%	2160
Vendor	2160	20%	2592

Student Claims

Claim	Verification	Judgment
"Total profit = 40%"	40% of 1800 = 720, SP should be 2520	WRONG
"Total profit = 44%"	44% of 1800 = 792, SP = 2592	CORRECT
"Carpenter earned more"	Carpenter profit = 360, Vendor profit = 432	WRONG
"Vendor earned more"	Vendor profit = 432, Carpenter profit = 360	CORRECT

Justification:

- Vendor's profit % is the same (20%) but on a higher CP (2160)
- Vendor earns more absolute profit (432 vs 360)
- Total profit% is NOT additive (20% + 20% ≠ 40%)

10 Evaluating Commission Situations

Scenario	Value	Commission %	Commission	Net	Seller Gets
A	1,000,000	5%	50,000	950,000	95%
B	1,000,000	3%	30,000	970,000	97%

Evaluation:

- Seller gets more in Scenario B (97% vs 95%)
- If you're the seller → choose lower commission
- If you're the broker → choose higher commission

Find Commission % from Given Data

Given	Analysis	Commission %
Broker gets 25,000 on 300,000	$25,000/300,000 \times 100$	8.33%
Seller receives 570,000, commission 30,000	$30,000/600,000 \times 100$	5%

11 Evaluating Word Problem Solutions

Problem 1: Stock of Fruits

Solution	Evaluation	Why?
Total CP = $60 \times 50 = 3000$	CORRECT	60 kg at Rs 50/kg
Total SP = $1400 + 900 + 500 + 200 = 3000$	CORRECT	All selling amounts
Profit = 0 (break-even)	CORRECT	CP = SP

Problem 2: Salary Increase

Solution	Evaluation	Why?
$25000 + 12\% = 28000$	WRONG (as written)	12% of 25000 = 3000, but "+12%" is not valid notation
$25000 \times 112/100 = 28000$	CORRECT	12% increase, calculated properly

Problem 3: School Population

Solution	Evaluation	Why?
$750 \times 20\% = 150$	CORRECT	20% of 750 = 150
150 students travel by bus	CORRECT	—

12 Judging Formula Choice

Problem Type	Correct Formula	Why?
Given CP and SP, find P%	$(SP - CP)/CP \times 100$	Profit% uses CP
Given CP and P%, find SP	$CP \times (100 + P)/100$	Add profit to CP
Given SP and P%, find CP	$SP \times 100/(100 + P)$	Reverse of above
Given MP and D%, find SP	$MP \times (100 - D)/100$	Subtract discount
Given SP and D%, find MP	$SP \times 100/(100 - D)$	Reverse of above
Given Value and C%, find Commission	$Value \times C/100$	Percentage of value
Given Commission and Value, find C%	$Commission/Value \times 100$	Commission as % of value

Judgment: Correct formula gives correct answer; wrong formula gives incorrect answer. Choosing the correct formula is essential!

13 Evaluating the Reasonableness of Profit/Loss

Data	Reasonable?	Why?
CP=100, Profit=200% → SP=300	YES	100% profit = 200, 200% profit = 300
CP=100, Profit=1000% → SP=1100	YES	Possible for extreme situations
CP=100, Profit=500% → SP=600	YES	$100 + 500 = 600$
CP=100, Loss=100% → SP=0	YES	Giving away for free
CP=100, Loss=200% → SP=-100	NO	Can't have loss > 100% (SP would be negative)
CP=100, Profit=50% → SP=125	NO	Should be 150 ($100 \times 1.5 = 150$)

14 Evaluation Checklist

When verifying an answer:

- Did I identify all given values?
- Did I choose the correct formula?
- Did I calculate correctly?
- Is the answer reasonable?
- Does it make sense in context?

When comparing options:

- What are all the options?
- What are the pros and cons of each?
- Which is better and why?
- Under what conditions would the other be better?

When judging student work:

- Did they use the correct formula?
- Did they use the correct base (CP, MP, Value)?
- Did they show their working?
- Is the final answer correct?
- If wrong, where did they make the mistake?

When evaluating a decision:

- What are the consequences?
- Is it profitable / loss-making?
- Is it fair to all parties?
- Would I make the same decision?

When checking reasonableness:

- Is $SP > CP$ when there should be profit?
- Is $SP < CP$ when there should be loss?
- Is $SP < MP$ when there should be discount?
- Is $Net < Value$ when there should be commission?

15 Self-Test — Evaluation Questions

No.	Question	Answer & Justification
1	CP=600, SP=750. Student says Profit=25%. Correct?	YES — $150/600 \times 100 = 25\%$
2	CP=800, SP=680. Student says Loss=20%. Correct?	NO — Loss=120, $120/800 \times 100 = 15\%$
3	Which is better: 10% of 500, or Rs 60 off on 500?	10% of 500 = 50 → Rs 60 off is better
4	Which is more profitable: 20% on CP=500, or 10% on CP=1000?	20% of 500=100, 10% of 1000=100 → Equal
5	Seller says 20% markup + 20% discount = no profit/loss. Correct?	NO — Discount on MP (larger) → Loss 4%
6	Student says Profit% = Profit/SP × 100. Correct?	NO — Should be Profit/CP × 100
7	Is 200% loss possible?	NO — SP would be negative, not possible
8	Commission 5% on 1,000,000 → Seller gets 950,000. Correct?	YES — $1,000,000 \times 95/100 = 950,000$
9	MP=1200, Discount=20% → SP=960. Correct?	YES — $1200 \times 80/100 = 960$
10	Which is better: 8% discount, or Rs 100 off on Rs 1500?	8% of 1500=120 > 100 → 8% discount is better

16 Quick Evaluation Guide

✓ CORRECT if:

- Formulas use correct base (CP for P/L, MP for discount, Value for commission)
- Calculations are accurate
- Answers are reasonable
- $SP > CP$ for profit, $SP < CP$ for loss

✗ INCORRECT if:

- Profit% uses SP as base
- Loss% uses SP as base
- Discount uses CP as base
- Commission uses CP as base
- Markup% = Discount% treated as break-even (causes loss!)

🔍 REASONABLE if:

- Profit% between 0 and 1000% (usually)
- Loss% between 0 and 100%
- Discount% between 0 and 100%
- Commission% between 0 and 100%

📊 BETTER DEAL:

- For customer → Lower SP
- For seller → Higher SP
- For investor → Higher Profit%
- For broker → Higher Commission%

Revision Tip: Always verify answers by working backwards. If SP=120 with 20% profit, then CP=100. Check: 20% of 100=20, 100+20=120 ✓. This confirms your answer is correct!